

DRY-RUN GUIDE | 13_expression.cpp
Source: cpp\1st_belt\13_expression.cpp

CORE BEHAVIOR

Reads four integers and evaluates $a+b*c-d/a$ using C++ precedence and integer division.

REPRESENTATIVE INPUT / CALL

Input: '10 2 3 9'.

STEP TRACE

Multiply first: $2*3=6$; integer divide: $9/10=0$; then $10+6-0=16$.

EXPECTED OUTPUT / RETURN

'16'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | You are given four integers a, b, c and d. Your task is to evaluate:
004 |
005 | result = a + b * c - d / a
006 |
007 | Use the standard mathematical operator precedence, where multiplication
008 | and division are performed before addition and subtraction. Perform
009 | integer arithmetic throughout, so d / a uses integer division.
010 |
011 | Input Format:
012 | A single line containing four space-separated integers a, b, c and d.
013 | */
014 |
015 | #include <iostream>
016 | using namespace std;
017 |
018 | int main() {
019 |     long long a, b, c, d;
020 |     cin >> a >> b >> c >> d;
021 |
022 |     long long result = a + b * c - d / a;
023 |
024 |     cout << result;
025 |
026 |     return 0;
027 | }
028 |
```